

Literature Status: the $1 < p < 2$ L_p Grothendieck Approximation Problem

Run fa_banach_001 19 July 2026

Status: `literature_already_answered` (conditional negative answer). This note records a later literature answer; it is not a new proof produced by the run.

Source question

Khot and Naor define, for an $n \times n$ input matrix $A = (a_{ij})$ with zero diagonal,

$$M_p(A) = \max_{\sum_{k=1}^n |t_k|^p \leq 1} \sum_{i=1}^n \sum_{j=1}^n a_{ij} t_i t_j.$$

In Section 5, printed page 26 (PDF page 27), they ask whether, for fixed $p \in (1, 2)$, $M_p(A)$ can be approximated in polynomial time within an $O(1)$ factor, and state that no hardness-of-approximation result was then known [1].

Identification of the answer

Bhattiprolu, Lee, and Naor prove in Section 8.1, Proposition 8.5(1), (4), and (5), of the full version of their STOC 2021 paper that the answer is negative under standard complexity assumptions [2]. For every fixed $q > 2$, assuming the Small Set Expansion Hypothesis and $P \neq NP$, there is no polynomial-time constant-factor approximation for the $2 \rightarrow q$ norm. Their Proposition 8.5 transfers this hardness first to the $q^* \rightarrow q$ norm and then to quadratic maximization over the ℓ_{q^*} unit ball, explicitly even for matrices with zero diagonal.

Given any fixed $p \in (1, 2)$, take $q = p^* = p/(p-1) > 2$. Then $q^* = p$, so the last statement is precisely constant-factor inapproximability of $M_p(A)$ in the zero-diagonal setting of the source question. The supporting paper also describes this as hardness for the ℓ_p Grothendieck problem that had been left open.

Scope limitation

This is a conditional negative answer. It rules out the requested algorithm assuming SSEH and $P \neq NP$; it does not establish an unconditional complexity separation. No stronger unconditional claim is made here.

References

- [1] S. Khot and A. Naor, *Grothendieck-type inequalities in combinatorial optimization*, Commun. Pure Appl. Math. 65 (2012), 992–1035; arXiv:1108.2464.
- [2] V. Bhattiprolu, E. Lee, and A. Naor, *A Framework for Quadratic Form Maximization over Convex Sets through Non-Convex Relaxations*, STOC 2021, 870–881, doi:10.1145/3406325.3451128; full draft dated 28 October 2022, author-hosted PDF.